Lecture 15 (October 27th)

Definition. (Time-dependent pertubation theory) In time-dependent pertubation theory, the Hamiltonian dependents on time as

$$\hat{H}(t) = \hat{H}^{(0)} + \lambda \hat{H}^{(1)}(t)$$

we assume that we know all information about $\hat{H}^{(0)}$, with $\{E_n^{(0)}, |\phi_n^{(0)}\rangle\}$. We can imagine like a laser being shot into a hydrogen atom and transitions occuring, similar to scattering and collisions. The goal is to of course solve

$$i\hbar \frac{\partial}{\partial t}|\psi(t)\rangle = \hat{H}(t)|\psi(t)\rangle$$

with $|\psi(t)\rangle = \sum_n c_n(t)e^{-iE_nt/\hbar}|\phi_n\rangle$. Substituting, we have a coupled ordinary differential equation with respect to $c_n(t)$. Our first goal will be to derive this. We have

$$\begin{aligned} i\hbar \sum_n \Big(\dot{c}_n(t) + \Big(-\frac{iE_n}{\hbar}\Big)c_n(t)\Big)e^{iE_nt/\hbar}|\phi_n\rangle \\ = \sum_n \Big(E_n^{(0)} + \lambda \hat{H}^{(1)}(t)\Big)c_n(t)e^{-iE_nt/\hbar}|\phi_n\rangle \end{aligned}$$

Applying $\langle\phi_m|$ on both sides,

$$i\hbar\dot{c}_m(t)e^{-iE_mt/\hbar} = \sum_n \lambda\langle\phi_m|\hat{H}^{(1)}(t)|\phi_n\rangle e^{-iE_nt/\hbar}c_n(t)$$

We conclude

$$i\hbar\frac{dc_m(t)}{dt} = \sum_n e^{i(E_m-E_n)t/\hbar}\lambda H_{mn}^{(1)}(t)c_n(t)$$

which is exact. In reality, in many cases, at $t = t_i$, only one state is occupied, say $|\phi_k\rangle$, and

$$c_n(t_i) = \delta_{nk}$$

Let's now approximate starting here.

$$i\hbar\frac{dc_{m\neq k}(t)}{dt} \approx e^{i(E_m-E_k)t/\hbar}H_{mk}^{(1)}(t)c_k(t)$$

is the first order approximation as we expect the initial energy to be accumulated at E_k. Solving,

$$c_{m\neq k}(t) - c_{m\neq k}(t_i) = \int_{t_i}^{t} dt'\, \frac{dc_{m\neq k}(t')}{dt'} = \int_{t_i}^{t} dt'\, \frac{1}{i\hbar}e^{i(E_m-E_k)t'/\hbar}V_{mk}(t')$$

Let's return to the original ansatz. We want the transition applitude $\langle\phi|\psi(t)\rangle = c_m(t)e^{-iE_m t}/\hbar$. Where the transition probability is given by $|\phi_m|\psi(t)\rangle|^2 = |c_m(t)|^2$. Notice how

$$\sum_m |c_m(t)|^2 = \sum_m \langle\psi(t)|\phi_m\rangle\langle\phi_m|\psi(t)\rangle = \langle\psi(t)|\psi(t)\rangle = 1$$

and the coefficients are normalised.

Example. (Harmonic oscillator) Consider the simple harmonic oscillator with $\lambda H^{(1)}(t') = qExe^{-(t')^2/\tau^2}$ where τ can be interpreted as pulse width. Take $t_i = -\infty$ and the ground state $n = 0$. Up to the first order, we have

$$c_{n\neq 0}(t) = \frac{qE}{i\hbar}\int_{-\infty}^{t} dt'\, e^{i\omega_{n0}t'}\langle n|x|0\rangle e^{-(t')^2/\tau^2}$$

where $\omega_{mn} = (E_m - E_n)/\hbar$. Also, notice that $\langle n|x|0\rangle$ is not zero only if $n = 1$. Computing the Gaussian integral,

$$P_m(\infty) = \pi\left(\frac{qE\tau}{\hbar}\right)^2 \times |\langle m|x|0\rangle|^2 \times e^{-\omega_{k0}^2\tau^2/4}$$

which is only nonzero for $m = 1$.

Example. (Other example) Let

$$\hat{H}^{(1)}(t) = \left(\hat{M}e^{i\omega t} + \hat{M}^\dagger e^{i\omega t}\right)$$

where we assume that $\hat{M} = \hat{M}^\dagger$ for simplicity. Then,

$$\hat{H}^{(1)}(t) = \hat{M}(e^{-i\omega t} + e^{i\omega t})$$

For first order, let $t_i = 0$ and $c_n(0) = \delta_{nk}$. We then have

$$\begin{aligned}
c_{m\neq k}(t) = &\int_0^t dt'\,\frac{1}{i\hbar}e^{i\omega_{mk}t'}\langle m|M|k\rangle\left(e^{-i\omega t'} + e^{i\omega t'}\right)\\
=&\frac{1}{i\hbar}M_{mk}\frac{e^{i(\omega_{mk}\mp\omega)t} - 1}{i(\omega_{mk}\mp\omega)}\\
=&\frac{1}{i\hbar}M_{mk}\frac{1}{i(\omega_{mk}\mp\omega)}e^{i(\omega_{mk}\mp\omega)t/2}\left(e^{i(\omega_{mk}\mp\omega)t/2} - e^{-i(\omega_{mk}\mp\omega)t/2}\right)\\
=&\frac{1}{i\hbar}M_{mk}\left(\frac{\sin((\omega_{mk}\mp\omega)t/2)}{(\omega_{mk}\mp\omega)t/2}\right)^2 e^{i(\omega_{mk}\pm\omega)t/2}
\end{aligned}$$

Lecture 16 (October 29th)

Theorem. (Fermi Golden Rule) Finishing last time's calculations,

$$|c_{m\leftarrow k}(t)|^2 = \frac{1}{\hbar^2}\Big[\frac{2-2\cos((\omega_{mk}-\omega)t)}{(\omega_{mk}-\omega)^2}|M_{mk}|^2 + \frac{2-2\cos((\omega_{mk}+\omega)t)}{(\omega_{mk}+\omega)^2}|M^\dagger_{mk}|^2\Big]$$

as $t\to\infty$. We can write the above in terms of sine squared and we have

$$|c_{m\leftarrow k}|^2 = \frac{1}{\hbar^2}\Big[\frac{\sin^2((\omega_{mk}-\omega)t/2)}{((\omega_{mk}-\omega)/2)^2}+\dots\Big]$$

where

$$\lim_{t\to\infty}\frac{\sin^2((\omega_{mk}-\omega)t/2)}{((\omega_{mk}-\omega)/2)^2} = 2\pi t\delta(\omega_{mk}-\omega)$$

Now,

$$|c_{m\leftarrow k}(t)|^2 = \frac{2\pi t}{\hbar^2}\Big[\delta(\omega_{mk}-\omega)|M_{mk}|^2+\delta(\omega_{mk}+\omega)|M^\dagger_{mk}|^2\Big]$$

recalling that $\omega_{mk}=(E_m-E_k)/\hbar$, we can write

$$|c_{m\leftarrow k}(t)|^2 = \frac{2\pi t}{\hbar}\Big[\delta(E_m-E_k-\hbar\omega)|M_{mk}|^2+\delta(E_m-E_k+\hbar\omega)|M^\dagger_{mk}|^2\Big]$$

which gives the transition probability per time, the transition rate, as

$$\Gamma = \Big(\frac{2\pi}{\hbar}\Big)\Big[\delta(E_m-E_k-\hbar\omega)|M_{mk}|^2+\delta(E_m-E_k+\hbar\omega)|M^\dagger_{mk}|^2\Big]$$

which is called the Fermi Golden Rule. Let's further investigate this and check units. We want it to be $[\Gamma]=T$. We see that, indeed,

$$[\Gamma] = \frac{1}{E\cdot T}\times\frac{1}{E}\times E^2 = \frac{1}{T}$$

In the general case,

$$\Gamma(i\to f) = \frac{2\pi}{\hbar}\sum\delta(\Sigma E)|M|^2$$

where the sum is over all final sttates.

Example. (Two-state harmonic pertubation) For simplicity, let $V_{11}=V_{22}=0$ for some perturbing Hamiltonian $\hat{V}$. Let

$$V_{12}=\gamma e^{i\omega t}$$

and let $\gamma>0$. Then, $V_{21}=\gamma^* e^{-i\omega t}$ where $\gamma^*=\gamma$. Up to first order, the time-dependent pertubation becomes

$$|\psi(t)\rangle = \sum_{n=1,2} c_n(t)e^{-iE_nt/\hbar}|\phi_n\rangle$$

The differential equation we need to solve is:

$$\begin{cases} i\hbar \dfrac{dc_1}{dt}(t) = V_{12} e^{i\omega_{12} t} c_2(t) \\ \\ i\hbar \dfrac{dc_2}{dt}(t) = V_{21} e^{i\omega_{21} t} c_1(t) \end{cases}$$

where $\omega_{12} = -\omega_{21}$. Setting $\omega - \omega_{21} = \Omega$, we can write

$$\begin{aligned} i\hbar \frac{dc_1}{dt}(t) =& \gamma e^{i\Omega t} c_2(t) \\ i\hbar \frac{dc_2}{t}(t) =& \gamma e^{-i\Omega t} c_1(t) \end{aligned}$$

Solving the coupled differential equations,

$$i\hbar \ddot{c}_2(t) = r\Big(-i\Omega e^{-i\Omega t} \frac{i\hbar \dot{c}_2(t)}{\gamma} e^{i\Omega t} + e^{-i\Omega t} \frac{\gamma}{i\hbar} e^{i\Omega t} c_2(t)\Big)$$

we obtain

$$\ddot{c}_2(t) = -i\Omega \dot{c}_2(t) - \frac{\gamma^2}{\hbar^2} c_2(t)$$

The general solution can be obtained through substituting $c_2(t) = Ae^{i\alpha t}$. We get the equation $-\alpha^2 = \alpha\Omega - \gamma^2/\hbar^2$ giving two real solutions for α.

$$\alpha_\pm = \frac{-\Omega \pm \varDelta}{2}$$

where $\varDelta = \sqrt{\Omega^2 + 5\gamma^2/\hbar^2}$. With the initial conditions $c_1(0) = 1$ and $c_2(0) = 0$,

$$c_2(t) = c_+ e^{i\alpha_+ t} + c_- e^{-i\alpha_- t}$$

with $c_+ = -c_-$ and

$$c_2(t) = c_+\Big(e^{i\alpha_+ t} - e^{i\alpha_- t}\Big) = (2i) c_+ e^{i(\alpha_+ + \alpha_-)t/2} \sin\Big(\frac{\varDelta t}{2}\Big)$$

where the condition $c_1(0) = 1$ gives $c_+ = -\gamma/\varDelta\hbar$. Thus,

$$|c_2(t)|^2 = \frac{\gamma^2/\hbar^2}{\dfrac{\gamma^2}{\hbar^2} + \Big(\dfrac{\omega - \omega_{21}}{2}\Big)^2} \sin^2\Big(\sqrt{\frac{\gamma^2}{\hbar^2} + \Big(\frac{\omega + \omega_{21}}{2}\Big)^2}\, t\Big)$$

which is known as the Rabi formula.

Definition. (Interaction representation) We see a middle ground representation of the Heisenberg and Schrodinger picture. The idea is simple – we write $\hat{H} = \hat{H}_0 + \lambda \hat{V}(t)$ and only let the time evolve via $\lambda\hat{V}(t)$. We express

$$|\alpha, t_0, t\rangle_I = e^{i\hat{H}_0 t/\hbar}|\alpha, t_0, t\rangle_S$$

and

$$|\alpha, t_0, t_0\rangle_S = e^{-iE_\alpha t_0/\hbar}|\alpha\rangle$$

which is important to remember. Then,

$$\hat{V}_I(t) = e^{iH_0 t/\hbar}V_s e^{-iH_0 t/\hbar}$$

Lecture 17 (November 3rd)

Definition. (Interaction picture) Let $\hat{H} = \hat{H}_0 + \lambda V(t)$. Let

$$|\alpha, t_0, t\rangle_I = e^{iH_0 t/\hbar}|\alpha, t_0, t\rangle_S$$

and

$$|\alpha, t_0, t_0\rangle_S = e^{-iE_\alpha t_0/\hbar}|\alpha\rangle$$

as the defintions of a interaction picture. We then have

$$i\hbar\frac{\partial}{\partial t}|\alpha, t_0, t\rangle_I = V_I(t)|\alpha, t_0, t\rangle_I$$

and

$$\frac{dV_I}{dt} = \frac{1}{i\hbar}[V_I, H_0] + e^{iH_0 t/\hbar}\frac{\partial V_S}{\partial t}e^{-iH_0 t/\hbar}$$

where $(H_0)_I = H_0$.

Proof.

$$\begin{aligned}
i\hbar\frac{\partial}{\partial t}|\alpha, t_0, t\rangle_I =& \Big(i\hbar\frac{\partial}{\partial t}e^{iH_0 t/\hbar}\Big)|\alpha, t_0, t\rangle_S + e^{iH_0 t/\hbar}\Big(i\hbar\frac{\partial}{\partial t}|\alpha, t_0, t\rangle_S\Big) \\
=& -H_0 e^{iH_0 t/\hbar}|\alpha, t_0, t\rangle_S + e^{iH_0 t/\hbar}\Big(H_0 + V(t)\Big)|\alpha, t_0, t\rangle_S \\
=& e^{iH_0 t/\hbar}V(t)|\alpha, t_0, t\rangle_S = e^{iH_0 t/\hbar}V(t)e^{-iH_0 t/\hbar}e^{iH_0 t/\hbar}|\alpha, t_0, t\rangle_S \\
=& V_I(t)|\alpha, t_0, t\rangle_I
\end{aligned}$$

□

Corollary. Previously, we have seen how

$$|\psi(t)\rangle_S = \sum_n c_n(t)e^{-iE_n t/\hbar}|n\rangle$$

Then,

$$
\begin{aligned}
|\psi(t)\rangle =& e^{iH_0t/\hbar}|\psi(t)\rangle_S = e^{iH_0t/\hbar}\sum_n c_n(t)e^{-iE_nt/\hbar}|n\rangle \\
=& \sum_n c_n(t)|n\rangle
\end{aligned}
$$

Corollary. In quantum field theory, the time evolution operator $U_I(t,t_0)$ resulted in

$$|\alpha, t_0, t\rangle_I = U_I(t,t_0)|\alpha, t_0, t_0\rangle_I$$

applying this to the time evolution equation for the interaction picture ket from the above,

$$i\hbar\frac{\partial}{\partial t}U_I(t,t_0) = V_I(t)U_I(t,t_0)$$

applying the pertubation here, we obtain the Dyson series or the Dyson expansion.

$$U_I^{(0)}(t,t_0) = \mathbf{1}$$

and

$$U_I(t,t_0) = \mathbf{1} + \Big(-\frac{i}{\hbar}\Big)\int_{t_0}^{t} dt'\, V_I(t')U_I(t',t_0)$$

from the differential equation gives

$$U_I^{(1)} = \mathbf{1} + \Big(-\frac{i}{\hbar}\Big)\int_{t_0}^{t} dt'\, V_I(t')$$

up to the first order. The second order is trivial as

$$
\begin{aligned}
U_I^{(2)} =& \mathbf{1} + \Big(-\frac{i}{\hbar}\Big)\int_{t_0}^{t} dt'\, V_I(t')U_I^{(1)}(t') \\
=& \mathbf{1} + \Big(-\frac{i}{\hbar}\Big)\int_{t_0}^{t} dt'\, V_I(t') + \Big(-\frac{i}{\hbar}\Big)^2\int_{t_0}^{t} dt'\, V_I(t')\int_{t_0}^{t'} dt''\, V_I(t'')
\end{aligned}
$$

Definition. (Berry phase) Berry phase is closely relevant to topology in physics. The geometry of quantum states is given by $|\Psi(R(t))\rangle \sim e^{i\theta}|\Psi(R(t))\rangle$ where $R(t)$ is a set of parameters (such as $\boldsymbol{R}$ for a line and S^2 for perhaps applied field direction). Imagine that we slowly vary the parameters, thus looking at an adiabatic transition. That is, we take $\Delta > \hbar\omega$ where Δ is the difference of energy levels. For instantaneous eigenkets, we have

$$H(R)|n(R)\rangle = E_n(R)|n(R)\rangle$$

Lecture 18 (November 10th)

Definition. We have previously defined

$$F_{\mu\nu}^{(n)} = i\Big[\sum_{\text{all},n}\langle\frac{\partial n}{\partial R^\mu}|n'\rangle\langle n'|\frac{\partial n}{\partial R^\nu}\rangle - \sum_{\text{all},n}\langle\frac{\partial n}{\partial R^\nu}|n'\rangle\langle n'|\frac{\partial n}{\partial R^\mu}\rangle\Big]$$

where

$$\langle n|\frac{\partial H}{\partial R}|n'\rangle = (E_n - E_n')\langle\frac{\partial n}{\partial R}|n'\rangle$$

for $n \neq n'$. What about the case where $n' = n$?

$$\langle\frac{\partial n}{\partial R^\mu}|n\rangle\langle n|\frac{\partial n}{\partial R^\nu}\rangle - \langle\frac{\partial n}{\partial R^\nu}|n\rangle\langle n|\frac{\partial n}{\partial R^\mu} = (ia)(ib) - (-ib)(-ia) = 0$$

Thus, we can change the summation index above from $n' \neq n$. We obtain

$$F_{\mu\nu}^{(n)} = i\sum_{n'\neq n}\Big[\frac{\langle n|\frac{\partial H}{\partial R^\mu}|n'\rangle\langle n'|\frac{\partial H}{\partial R^\nu}|n\rangle}{(E_n - E_{n'})^2} - (\cdots)^*\Big]$$

This is the most direct formula. Summing both sides over with respect to n, we can then change n with n' and find that the summation over n is zero. for convenience, let's define a 2×2 Herimian matrix that is traceless iwth basis O_x, O_y, O_z as the basis.

$$H = h_x O_x + h_y O_y + h_z O_z = \begin{pmatrix} h_z & h_x - ih_y \\ h_x + ih_y & -h_z \end{pmatrix}$$

We take $\vec{h} = (h_x, h_y, h_z) \in \boldsymbol{R}$ to be real and we slowly vary the parameters. The eigenvalues are $-h$ and h where $h = \sqrt{h_x^2 + h_y^2 + h_z^2}$. Let's now express everything in spherical coordinates $h_x = \sin\theta\cos\phi$, $h_y = h\sin\theta\sin\phi$, and $h_z = h\cos\theta$. Consider the eigenstates $|-\rangle$ and $|+\rangle$. Recall that there is phase ambiguity in making the choices

$$|-\rangle = \begin{pmatrix} \sin(\theta/2)e^{-i\phi} \\ -\cos(\theta/2) \end{pmatrix}$$

and

$$|+\rangle = \begin{pmatrix} \cos(\theta/2)e^{-i\phi} \\ \sin(\theta/2) \end{pmatrix}$$

The nontrivialness of the parameter space's topology leads to non-uniqueness of global eigenvectors!

Corollary. (In the test) Let's calculate the Berry phase. We can easily check that $\langle\pm|\vec{O}|\pm\rangle = \pm\vec{h}/h$. For example, $\langle -|O_z|-\rangle = \sum^2(\theta/2) - \cos^2(\theta/2) = -\cos^2\theta = -h_z/h$. Notice that

$$A_\theta^{(-)} = \langle -|i\frac{\partial}{\partial\theta}|-\rangle = 0$$

and

$$A_\phi^{(-)} = \langle -|i\frac{\partial}{\partial\phi}|-\rangle = \sin^2(\theta/2) = \frac{1}{2}(1-\cos\theta)$$

We will in the far future learn that 1/2 actually denotes spin. Recall that the field strength was defined as

$$F_{\theta\phi}^{(-)} = \partial_\theta A_\phi - \partial_\phi A_\theta = \frac{1}{2}\sin\theta$$

In the language of differential forms,

$$F^{(-)} = \frac{1}{2}\sin\theta\, d\theta\wedge d\phi$$

which we can rewrite as

$$F^{(-)}\frac{S}{h^2}(h^2\sin\theta\, d\theta\wedge d\phi)$$

Integrating over the sphere,

$$\int_{S^2} F^{(-)}\, d\vec{S} = \frac{s}{h^2}(4\pi h^2) = 4\pi s$$

We find $\int \vec{B}\cdot d\vec{S} = \int_{S^2} F^{(-)}$ and

$$\vec{B} = \frac{s}{h^2}\hat{h}$$

where the magnetic sitting at $h=0$ is unique. By the way, check $F^{(+)} = -F^{(-)}$.

Example. We try to obtain the above by this time using the direct formula. Recall that $F_{xy} = B_z$. Using

$$F_{\mu\nu}^{(-,+)} = i\sum_{m\neq n}\left(\left\langle n\middle|\frac{\partial H}{\partial R^\mu}\middle|m\right\rangle\left\langle m\middle|\frac{\partial H}{\partial R^\mu}\middle|n\right\rangle - (\cdots)^*\right)$$

and

$$\frac{\partial H}{\partial h_x} = O_x \qquad \& \qquad \frac{\partial H}{\partial h_y} = O_y$$

At the end of the day,

$$F_{xy} = \frac{1}{2}\frac{h_z}{h^3}$$

which I want you to check!

Corollary. Let $H = \vec{p}^2/2m + V(\vec{x},t)$. What is $\vec{A}$? Recall that $ma = \vec{F} = q(\vec{E}+\vec{v}\times\vec{B})$. We find that

$$H = \frac{(\vec{p}-q\vec{A})^2}{2m} + q\phi$$

which would be memorised. Note, $\vec{B} = \nabla\times\vec{A}$ and $\vec{E} = -\nabla\phi - \partial\vec{A}/\partial t$. We later check that the Hamilton's equations of motion work from this.

Lecture 19 (November 12th)

Recall. We want a quantum Hamiltonian that incorporates electric and magnetic fields. We've previously written:

$$H = \frac{1}{2m}(\vec{p} - q\vec{A})^2 + q\phi$$

We know the Hamiltonian equations of motion as

$$\begin{cases} \dfrac{dx_i}{dt} = \dfrac{\partial H}{\partial p_i} \qquad \longrightarrow \qquad \dfrac{dx_i}{dt} = \dfrac{1}{m}(p_i - qA_i) \\[2em] \dfrac{dp_i}{dt} = -\dfrac{\partial H}{\partial x_i} \qquad \longrightarrow \qquad \dfrac{dp_i}{dt} = -\dfrac{1}{m}\sum_j (p_j - qA_j)(-q)\Big(\dfrac{\partial A_j}{\partial x_i}\Big) + q\dfrac{\partial \phi}{\partial x_i} \end{cases}$$

we of course expect, $\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$. Further manipulating the above,

$$\begin{aligned} m\frac{d^2 x_i}{dt^2} =& \Big(\frac{dp_i}{dt}\Big) - q\frac{dA_i}{dt} = \Big(\frac{dp_i}{dt}\Big) - q\Big(\sum_j \frac{\partial A_i}{\partial x_j}\frac{dx_j}{dt} + \frac{\partial A_i}{\partial t}\Big) \\ =& q\Big(-\frac{\partial A_i}{\partial t} - \frac{\partial \phi}{\partial x_i}\Big) - q\sum_j \frac{\partial A_i}{\partial x_j}\frac{dx_j}{dt} + q\sum_j \frac{\partial A_j}{\partial x_i}\frac{dx_j}{dt} \\ =& qE_i + \sum_j q\frac{dx_j}{dt}\Big(\frac{\partial A_j}{\partial x_i} - \frac{\partial A_i}{\partial x_j}\Big) \\ =& q(E_i + [\vec{v} \times \vec{B}]_i) \end{aligned}$$

which properly motivates our Hamiltonian above. Noticing the dependencies,

$$H(x,p) = \frac{1}{2m}(\vec{p} - q\vec{A}(\vec{x},t))^2 + q\phi(\vec{x},t)$$

we write the Schrodinger equation

$$i\hbar\frac{\partial}{\partial t}\psi(\vec{x},t) = H(\vec{x},\vec{p})$$

Recall that for electric and magnetic potential, $U(1)$ gauge symmetry exists, and

$$\vec{A} \to \vec{A} - \nabla\Lambda \qquad \& \qquad \phi \to \phi + \frac{\partial \Lambda}{\partial t}$$

should leave physical results invariant. Let's assume that

$$\psi' = e^{-iq\Lambda/\hbar}\psi$$

Substituting,

$$i\hbar\frac{\partial}{\partial t}\psi = i\hbar\frac{\partial}{\partial t}\Big(e^{iq\Lambda/\hbar}\psi'\Big) = e^{iq\Lambda/\hbar}\Big[-q\dot{\Lambda}\psi' + i\hbar\frac{\partial}{\partial t}\psi\Big]$$

and

$$q\phi e^{iq\Lambda/\hbar}\psi' = e^{iq\Lambda/\hbar}\Big[q\phi + q\frac{d\Lambda}{dt}\Big]\psi' = q\Big(\phi + \frac{\partial\Lambda}{\partial t}\Big)\psi'$$

therefore we see that ψ' is the propor transformed solution particularly for the ϕ transformation. Now,

$$\begin{aligned}(\vec{p} - q\vec{A})e^{iq\Lambda/\hbar}\psi' =& i\hbar\frac{\partial}{\partial\vec{x}}\Big(e^{iq\Lambda/\hbar}\psi'\Big)^2 - q\vec{A}e^{iq\Lambda/\hbar}\psi' \\ =& \Big(q\frac{\partial\Lambda}{\partial\vec{x}}\psi'\Big)e^{iq\Lambda} + (-i\hbar)\frac{\partial\psi'}{\partial\vec{x}}e^{-iq\Lambda/\hbar} - q\vec{A}e^{iq\Lambda/\hbar}\psi' \\ =& e^{iq\Lambda/\hbar}\Big[-i\hbar\frac{\partial}{\partial\vec{x}} - q\Big(\vec{A} - \frac{\partial\Lambda}{\partial\vec{x}}\Big)\Big]\psi' \\ (\vec{p} - q\vec{A})^2 e^{iq\Lambda/\hbar}\psi' =& \Big(-i\hbar\frac{\partial}{\partial\vec{x}} - q\Big(\vec{A} - \frac{\partial\Lambda}{\partial\vec{x}}\Big)\Big)^2\psi'\end{aligned}$$

Dividing by $2m$, we can see that the ψ' is a solution for the transformed gauge fields. This tells us that we can choose specific gauges that satisfy

$$\nabla\cdot\vec{A} = 0 \qquad \& \qquad \nabla\cdot\vec{A} + \frac{1}{c^2}\frac{\partial\phi}{\partial t} = 0$$

each called the Coulumb gauge and Lorentz gauge. We also need a vector potential $\vec{A}$, and two popular choices are the Landau gauge and the radial gauge:

$$\vec{A} = (0, Bx, 0) \qquad \& \qquad \vec{A} = \Big(-\frac{yB}{2}, \frac{Bx}{2}, 0\Big)$$

These two are actually gauge equivalent, meaning that there is a Λ such that the two gauges have a difference of $\nabla\Lambda$. j

Lemma. (Landau levels) We now solve the Schrodinger equation for the above two gauges that give the same magnetic field. Let $\vec{B} = B\hat{z}$ and $\phi = 0$. For an electron,

$$\begin{aligned}H\psi =& \frac{1}{2m_e}(-i\hbar\nabla - (-e)\vec{A})^2\psi \\ =& \frac{(-i\hbar\nabla)^2}{2m_e}\psi + \frac{1}{2m_e}(-i\hbar\nabla)\cdot(e\vec{A}\psi) + \frac{1}{2m_e}e\vec{A}\cdot(-i\hbar\nabla\psi) + \frac{e^2}{2m_e}\vec{A}\cdot\vec{A}\psi\end{aligned}$$

the second term becomes

$$\frac{1}{2m_e}(-i\hbar)\Big[(e\nabla\cdot A)\psi + e\vec{A}\cdot\nabla\psi\Big]$$

and due to our gauge term,

$$H\psi = \frac{(-i\hbar\nabla)^2}{2m_e}\psi + \frac{1}{m_e}e\vec{A}\cdot(-i\hbar\nabla\psi) + \frac{e^2}{2m_e}\vec{A}\cdot\vec{A}\psi$$

The middle term now becomes

$$\frac{1}{m_e}e\vec{A}\cdot(-i\hbar\nabla\psi) = \frac{e}{m_e}\frac{\vec{B}\cdot\vec{r}}{2}\times\vec{p}\psi = \frac{e}{m_e}\vec{B}\cdot\vec{L}\psi = \frac{e}{2m_e}BL_z\psi$$

and $\vec{A}\cdot\vec{A} = B^2(x^2+y^2)/4$ in the last term. This motivates us to use cylindrical coordinates, and

$$H\psi = \frac{(-i\hbar\nabla)^2}{2m_e}\psi + \frac{e}{2m_e}BL_z\psi + \frac{e^2B^2}{8m_e}\rho^2\psi$$

We can now take the ansatz

$$\psi(\rho,\phi,z) = e^{ik_z z}e^{im\phi}u_m(\rho)$$

and we get

$$\frac{d^2u_m}{d\rho^2} + \frac{1}{\rho}\frac{du_m}{d\rho} - \frac{m^2}{\rho^2}u_m - \frac{e^2B^2}{4\hbar^2}\rho^2 u_m + \Big(\frac{2m_eE}{\hbar^2} - \frac{eB\hbar m}{\hbar^2} - k_z^2\Big)u_m = 0$$

employing dimensionless $x = \sqrt{eB/2\hbar}\times\rho$ and $l_B = \sqrt{\hbar/eB}$ (magnetic length) that has the units of length. You should definitely check this and memorize this! We can also introduce the dimensionless

$$\lambda = \frac{4m_e}{\hbar}\Big(E - \frac{\hbar^2k_z^2}{2m}\Big) - 2m$$

and we have

$$u_m'' + \frac{1}{x}u_m' - \frac{m^2}{x^2}u_m - x^2u_m + \lambda u_m = 0$$

The next plan is to find the asymptotic solution for $x\to\infty$ and $x\to 0$ which will be done next class.

Theorem. We need to solve

$$u_m'' + \frac{1}{x}u_m' - \frac{m^2}{x^2}u_m - x^2u_m + \lambda u_m = 0$$

At $x\to\infty$, $u_m'' - x^2u_m \sim 0$ and $u_m \sim e^{-x^2/2}$ and $x\to 0$ gives $u_m'' + u_m'/x - |m|^2u_m/x^2 \sim 0$ giving $u_m \sim x^{|m|}$. Substituting $u(x) = e^{-x^2/2}x^{|m|}G(x)$,

$$G'' + \Big(\frac{2|m|+1}{x} - 2x\Big)G' + (\lambda - 2 - 2|m|)G = 0$$

taking $y = x^2$, we have Laguere equation

$$H''(y) + \Big(\frac{2l+2}{\rho} - 1\Big)H' + \frac{n_r}{\rho}H = 0$$

for integers n_r. The energy levels are

$$E(n_r,m,k_z) = \frac{\hbar^2k_z^2}{2m} + \frac{eB\hbar}{m_e}(n_r + \frac{1}{2} + \frac{|m|+m}{2})$$

with eigenfunctions

$$G(x^2) = L_{n_r}^{|m|}(x^2)$$

and $h\omega_c = eB\hbar/m_e$. The degeneracy factor becomes approximately

$$N_{\text{dege}} = \frac{\text{Area}}{2\pi l_B^2}$$

We must know why does this degeneracy occurs at the ground state $n_r = 0$. Memorise this formula.

Definition. (Landau gauge) For a vector potential gauge $\vec{A} = (0, Bx, 0)$, consider $H = (\vec{p} - (-e)\vec{A})^2$.

$$H = \frac{p_x^2}{2m_e} + \frac{e^2B^2x^2 + 2eBxp_y}{2m_e} + \frac{p_y^2}{2m_e} + \frac{p_z^2}{2m_e}$$

this tells us that $[H, p_y] = [H, p_z] = 0$ and that

$$\psi(x, y, z) = v(x)e^{ik_y y}e^{ik_z z}$$

Substituting,

$$\Big(-\frac{\hbar^2}{2m_e}\frac{d^2}{dx^2} + \frac{1}{2m_e}e^2B^2(x + x_0)^2\Big)v(x) = \Big(E - \frac{\hbar^2k_z^2}{2m_e}\Big)v(x)$$

where $x_0 = \hbar k_y/eB$. Setting $s = x + x_0$,

$$\Big(-\frac{\hbar^2}{2m_e}\frac{d^2}{ds^2} + \frac{1}{2m_e}e^2B^2s^2\Big)v(s - x_0) = \Big(E - \frac{\hbar^2k_z^2}{2m_e}\Big)v(s - x_0)$$

and

$$v(s - x_0) = e^{-m_e\omega_c s^2/\hbar}H_n(y)$$

with $y = s\sqrt{m_e\omega_c/\hbar}$ and $e^2B^2/2m_e = m_e\omega_c^2/2$. The eigenvalues are

$$E_n = \hbar\omega_c\Big(n + \frac{1}{2}\Big) + \frac{\hbar^2k_z^2}{2m_e}$$

and the structure is similar to the above. Taking the length of the Hall bar to be L_y and L_x, we have

$$k_y = \frac{2\pi}{L_y}n^*$$

and ecetera, giving the range

$$0 \leq \frac{2\pi\hbar}{L_y}\frac{1}{eB}n^* \leq L_x$$

and

$$0 \leq n^* \leq \frac{L_xL_y}{2\pi eB/\hbar} = \frac{\text{Area}}{2\pi l_B^2}$$

Lecture 21 (November 19th)

Definition. (Aharonov-Bohm effect) Consider a vector potential $\vec{A}$ and the generated magnetic field $\vec{B} = \nabla \times \vec{A}$. In the quantum world, the vector potential is more fundamental in some sense than the emergent field. Consider a solinoid of radius a with a homogeneous magnetic field with strength B_0. What is the vector potential? Well, the answer is, from the homework,

$$\vec{A} = \begin{cases} A_\phi(\rho < a) \sim \rho B_0/z \\ A_\phi(\rho > a) \sim \Phi_B/2\pi\rho \end{cases}$$

with $\Phi_B = B_0 \pi a^2$. Let's try solving Schrodinger's equation outside the solenoid with $\vec{B} = 0$ by trying to set $\vec{A}' = 0$ outside.

$$\vec{A}' = 0 = \vec{A} - \left(\hat{\rho}\frac{\partial}{\partial \phi} + \hat{\phi}\frac{1}{\rho}\frac{\partial}{\partial \phi} + \hat{z}\frac{\partial}{\partial z}\right)\Lambda$$

we find that we need $\Lambda = \Phi\phi/2\pi$. We recall that

$$\psi' = \exp\left[\frac{-iq\Lambda}{\hbar}\right]\psi = \exp\left[\frac{ie}{\hbar}\frac{\Phi}{2\pi}\phi\right]\psi = \exp\left[\frac{ie}{\hbar}\int \vec{A}\cdot d\vec{r}\right]\psi$$

as $d\vec{r} = \rho\, d\vec{\phi}$. We of course need our single-valuedness condition,

$$\psi(\phi + 2\pi) = \psi(\phi)$$

but

$$\psi(\phi + 2\pi) = e^{ie\Phi/\hbar}\psi(\phi)$$

and the phase of the wavefunction changes. For a two slit interference experiment,

$$\begin{aligned} \psi_1 + \psi_2 =& S_1 e^{iP_1} + S_2 e^{iP_2} \\ =& e^{iP_1}(S_1 + S_2 e^{iP_2 - iP_1}) \end{aligned}$$

where the phase difference is affected by the nontrivial integral

$$\oint \vec{A}\cdot d\vec{r}$$

and the interference pattern changes. A related problem will be in the test!

Theorem. (Radiation theory) First of all, define

$$H = \frac{(\vec{p} - (-e)\vec{A})^2}{2m_e}$$

where $\vec{A}$ is given as the vector potential of light. The most widely used is the coulomb gauge, with $\nabla \cdot \vec{A} = 0$. In vaccum, $\Phi = 0$, and we simply have

$$\vec{A} = \nabla \times \vec{A} \qquad \& \qquad \vec{E} = -\frac{\partial \vec{A}}{\partial t}$$

Second of all, we must discuss whether our system is semi-classical or quantum. For a true quantum picture, we need

$$a^r(k) \qquad \& \qquad a^r(k)^\dagger \qquad [a^r(k), a^r(k)^\dagger] = 1$$

The field becomes, approximately

$$\vec{A}(x,t) = \sum_k \sum_{r=1}^{2} \left[\vec{\epsilon}^{\,r}(k) a^r(k) e^{-i\omega t + ikx} + \vec{\epsilon}^{\,r}(k)^\dagger a^r(k)^\dagger e^{i\omega t - ikx} \right]$$

Most simply, let's consider monochromatic light with just a single ω. Noting that

$$N\hbar\omega = \int_V d^3\vec{x} \left(\frac{\epsilon_0}{2} \vec{E} \cdot \vec{E} + \frac{\mu_0}{2} \vec{B} \cdot \vec{B} \right)$$

and $\vec{E}$ and $\vec{B}$ from above, we can determine the normalisation of the ansatz above.

$$2 \int d^3\vec{x} \, \frac{\epsilon_0}{2} \langle \vec{E} \cdot \vec{E} \rangle_{\text{avg}} = 2\omega^2 V \epsilon_0 C^2$$

which gives

$$C = \sqrt{\frac{N\hbar}{2\epsilon_0 \omega V}}$$

Lecture 22 (November 24th)

Definition. Recall that under the gauge choice of $\nabla \cdot A = 0$, we can express the vector potential in the form

$$\vec{A}(\vec{x},t) = \sum_{\vec{k}} \sum_{\lambda} c \left[\hat{\epsilon}^{(\lambda)} A^{(\lambda)}(\vec{k}) e^{i\vec{k}\cdot x - i\omega t} + \hat{\epsilon}^{(\lambda)\,*} A^{\dagger\,(\lambda)}(k) e^{-i\vec{k}\cdot\vec{x} - i\omega t} \right]$$

The ladder operators are suppose to satisfy

$$[A^{(\lambda)}(\vec{k}), A^{\dagger\,(\lambda')}(\vec{k}')] = \delta^{\lambda\lambda'} \delta(\vec{k} - \vec{k}')$$

We expect

$$A^{\dagger\,(\lambda)}(\vec{k}) |N_{\lambda k}\rangle = \sqrt{N_{\lambda k} + 1} |N_{\lambda k} + 1\rangle$$

and etcetera. Recall that

$$H_{cl} = \int d^3x \left(\frac{\epsilon_0}{2} E^2 + \frac{1}{2\mu_0} B^2 \right)$$

and

$$\langle H_{cl} \rangle_t = 2 \cdot \frac{\epsilon_0}{2} \int d^3x \, \langle E^2 \rangle = N \times \hbar\omega$$

As $\vec{E} = -\partial \vec{A}/\partial t$,

$$\langle H_{cl} \rangle = \epsilon_0 c^2 V \omega^2 (AA^\dagger + A^\dagger A) = \epsilon_0 c^2 V \omega^2 \Big(A^\dagger A + \frac{1}{2} \Big) = N\hbar\omega$$

We find

$$c^2 = \frac{\hbar}{2V\omega\epsilon_0}$$

The ladder operators are expected to evolve as

$$a(t) = a(0)e^{-i\omega t} \qquad \& \qquad a^\dagger(t) = a^\dagger(0)e^{i\omega t}$$

in the Heisenberg picure. The vector potential in the Fock spcae should be definitely memorised.

Definition. (Radiative decay) We see one of the most prominent examples of the above construction. Recall that the Fermi golden rule was given as

$$\Gamma = \frac{2\pi}{\hbar} \sum_f |\langle f|M|i\rangle|^2 \delta(\Sigma E)$$

In the above, we take $M = e\vec{A}\cdot\vec{p}/m_e$.The phase space integral (summation over f) produces a factor V, which is cancelled with the factor in $\vec{A}$. The spin sector is namely,

$$\langle 1s| \frac{e}{m_e} \sqrt{\frac{\hbar}{2\epsilon_0\omega V}} e^{-i\vec{k}\cdot\vec{x}} \hat{\epsilon}^{(\lambda)\,*} \cdot \vec{p} \, |2p\rangle$$

For linear polarisation, $\hat{\epsilon}^* = \hat{\epsilon}$. Taking out the important parts,

$$\langle 1s|(1 - i\vec{k}\cdot\vec{x})\hat{\epsilon}\cdot\vec{p}|2p\rangle$$

Let's consider the 0th order first (electric dipole).

$$\begin{aligned}
\langle 1s|\hat{\epsilon}\cdot\vec{p}|2p\rangle =& \hat{\epsilon}\cdot\langle 1s|\vec{p}|2p\rangle = \hat{\epsilon}\cdot\langle 1s|m_e \frac{d\vec{x}}{dt}|2p\rangle = \hat{\epsilon}\cdot\langle 1s|m_e \frac{i}{\hbar}[H_0, \vec{x}]|2p\rangle \\
=& \frac{im_e}{\hbar}\vec{\epsilon}\cdot\langle 1s|H_0\vec{x} - \vec{x}H_0|2p\rangle = (E_f - E_i)\Big(\frac{im_e}{\hbar}\Big)\hat{\epsilon}\cdot\langle 1s|\vec{x}|2p\rangle
\end{aligned}$$

where the last term is essentially $\vec{p}$. Due to the Kronecker delta, the first round bracket becomes $-\hbar\omega$.

Lecture 23 (November 26th)

Theorem. We continue to calculate the decay rate from a $2p$ state to a $1s$ state emiting a photon. We use Fermi's golden rule, where

$$\Gamma = \frac{2\pi}{\hbar} \sum_{\text{final}} |\langle f|M|i\rangle|^2 \delta(\Sigma E)$$

We've expressed all states as a tensor product of the atomic wavefunction and the photon wavefunction, and 2nd quantised $\vec{A}$. The transition Hamiltonian is given as

$$M = \frac{e}{m_e} \vec{A} \cdot \vec{p}$$

that came from the first order of the original Hamiltonian. To compute the inner product of the initial and final atomic states, we needed a particular trick, where we needed to calculate

$$\langle \phi_f | \vec{x} \cdot \hat{\epsilon} | \phi_i \rangle = \int d\Omega \, Y^*_{l_f m_f} (\hat{\epsilon} \cdot \hat{r}) Y_{l_i m_i} = \int d\Omega \, Y^*_{l_f m_f} (\epsilon_x r_x + \epsilon_y r_y + \epsilon_z r_z) Y_{l_i m_i}$$

where

$$\epsilon_x r_x + \epsilon_y r_y + \epsilon_z r_z = \sqrt{\frac{4\pi}{3}} \left(\epsilon_z Y_{10} + \frac{-\epsilon_x + i\epsilon_y}{\sqrt{2}} Y_{11} + \frac{\epsilon_x + i\epsilon_y}{\sqrt{2}} Y_{1,-1} \right)$$

noting that

$$(Y_{lm})^* = (-1)^m Y_{l,-m} \qquad \& \qquad \mathcal{P} Y_{lm} \to (-1)^l Y_{lm}$$

where $\mathcal{P}$ is the parity operator. For simplicity, let's assume $\vec{R} \,||\, \hat{z}$. $\epsilon_z = 0$ and $m_f - m_i = \pm 1$. We can then compute the above using

$$\int d\Omega \, Y^*_{l'm'} Y_{lm} = \delta_{ll'} \delta_{mm'}$$

Through all this, we get to obtain

$$d\Gamma = \frac{1}{3} \sum_{m_i = \pm 1, 0} d\Gamma_m \propto (\epsilon_x^2 + \epsilon_y^2 + \epsilon_z^2)$$

and numerically, $\Gamma(2p \to 1s) = 0.62 \times 10^6$ per second.

Example. To this point, we have considered no change in spin, $\Delta S = 0$. However, $\Delta S \neq 0$ is indeed possible, where

$$\frac{eg}{m_e c} \vec{S} \cdot \vec{B}$$

and $\vec{B} = \nabla \times \vec{A}$. Consider the reaction:

$$\gamma + d \to p + n$$

where d is in the ${}^3S_1$ triplet state and $p + n$ is in the ${}^1S_0$ singlet state. Thus, $\Delta S = 1$. We find

$$\langle f|(g_p\vec{S}_p + g_n\vec{S}_n)\cdot\vec{B}|i\rangle \neq 0$$

in the homework problems.

Theorem. (No clone theorem) Consider an arbitrary qubit $|x\rangle \in \boldsymbol{C}^2$. Consider a clone map (a copy map):

$$G : \boldsymbol{C}^2 \times \boldsymbol{C}^2 \to \boldsymbol{C}^2 \times \boldsymbol{C}^2$$

which is a unitary operator. It would send

$$|x\rangle \otimes |0\rangle \to |x\rangle \otimes |x\rangle$$

where we emphasize that we do not know $|x\rangle$. The no clone theorem suggests that there doesn't exist such a G for a general $|x\rangle$.

Lecture 24 (December 1st)

Theorem. (No-clone theorem) Consider an arbitrary qubit $|x\rangle \in \boldsymbol{C}^2$. Consider a clone map (a copy map):

$$G : \boldsymbol{C}^2 \times \boldsymbol{C}^2 \to \boldsymbol{C}^2 \times \boldsymbol{C}^2$$

which is a unitary operator. I would send

$$|x\rangle \otimes |0\rangle \to |x\rangle \otimes |x\rangle$$

No such unitary map exists.

Proof. Let $|x\rangle = \alpha|0\rangle + \beta|0\rangle$ which is normalised, meaning $|\alpha|^2 + |\beta|^2 = 1$. As the map is linear,

$$\begin{aligned} G(\alpha|0,0\rangle + \beta|1,0\rangle) =& \alpha G(|0,0\rangle) + \beta G(|1,0\rangle) = \alpha|0,0\rangle + \beta|1,1,\rangle \\ =& \alpha^2|0,0\rangle + \beta\alpha|1,0\rangle + \alpha\beta|0,1\rangle + \beta^2|1,1\rangle \end{aligned}$$

As one of the parameters are zero, only two possible pairs (α, β) are possible. These are $|0\rangle$ and $|1\rangle$, telling us that the only qubits that are capable of copying are the basis qubits. □

Theorem. (Entanglement entropy) We would like to know whether a vector inside $\mathcal{H}_A \otimes \mathcal{H}_B$ is entangled or disentangled. Let

$$|\Psi\rangle = \sum_{i,j} M_{ij}|i\rangle_A \otimes |j\rangle_B$$

where $|i\rangle_A$ and $|j\rangle_B$ are orthonormal basis elements. The density matrix is given as

$$\rho_{|\Psi\rangle} = |\Psi\rangle\langle\Psi| = \sum_{i,j} M_{ij}|i\rangle_A \otimes |j\rangle_B \sum_{i',j'} M^*_{i',j'}\langle i'|_A \otimes \langle j'|_B$$

Let's partially trace the above, obtaining $\sum_k \langle k|_B \rho_{|\Psi\rangle} |k\rangle_B$.

$$\sum_k \langle k|_B \rho_{|\Psi\rangle} |k\rangle_B = \sum_i M_{ik} |i\rangle_A \sum_{i'} M^*_{i'k} \langle i'| = \sum_{i,i',k} M_{ik} M^*_{i'k} |i\rangle_A \langle i'|_A = \sum_{i,i'} (MM^\dagger)_{ii'} |i\rangle_A \langle i'|_A$$

When we partially trace, we find a density matrix with respect to $|i\rangle_A$, and we call $\rho_A = MM^\dagger$ and $\rho_B = M^\dagger M$ partial density matrices. Recall that for a density operator $\hat{\rho}$, the von Neumann entropy is defined as

$$S_N = -\mathrm{Tr}(\hat{\rho} \ln \hat{\rho})$$

Now, the entanglement entropy is defined as the von Neumann entropy of ρ_A and ρ_B. We show that

$$S_E = -\mathrm{Tr}(\rho_A \ln \rho_A) = -\mathrm{Tr}(\rho_B \ln \rho_B)$$

Observe that

$$-\mathrm{Tr}(\rho_A \ln \rho_A) = -\sum_{S_l > 0} S_l^2 \ln S_l^2 = -\mathrm{Tr}(\rho_B \ln \rho_B)$$

Example. Let $|\Psi\rangle = |\uparrow\rangle_A \otimes |\uparrow\rangle_B$ (which is obviouisly disentangled). We see that

$$M = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = M^\dagger$$

and we have

$$MM^\dagger = M^\dagger M = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

and we find that $S_1 = 1$. This gives

$$S_E = -1 \ln 1 = 0$$

and we can notice how $S_E = 0$ when the system is disentangled and $S_E > 0$ when the system is.

Example. Let $|\Psi\rangle = (|\uparrow\rangle_A \otimes |\downarrow\rangle_B - |\downarrow\rangle_A \otimes |\uparrow\rangle_B)/\sqrt{2}$. We find

$$M = \begin{pmatrix} 0 & 1/\sqrt{2} \\ -1/\sqrt{2} & 0 \end{pmatrix} \qquad \& \qquad M^\dagger = \begin{pmatrix} 0 & -1/\sqrt{2} \\ 1/\sqrt{2} & 0 \end{pmatrix}$$

giving

$$MM^\dagger = \begin{pmatrix} 1/2 & 0 \\ 0 & 1/2 \end{pmatrix}$$

We find

$$S_E = -\frac{1}{4} \ln \frac{1}{4} - \frac{1}{4} \ln \frac{1}{4} = \ln 2 \neq 0$$

Definition. (Differential scattering cross-section) In a scattering experiment, we define the differential scattering cross-section as the following:

$$d\sigma = \frac{\text{number of particles detected at } d\Omega \text{ per unit time}}{\text{incoming flux of particle}}$$

The unit is given as $[d\sigma] = L^2$. We now always compute things in the center of mass frame. Recall that the probability flux was given as

$$j = \frac{\hbar}{2im}(\psi^*\nabla\psi - \psi\nabla\psi^*) = \frac{1}{V}\frac{\hbar\vec{k}}{m} = \rho\vec{v}$$

in the case of $\psi = e^{ikx}/\sqrt{V}$.

Lecture 25 (December 3rd)

Remark. We have previously defined the differential cross section as the number density (over solid angle) per time divided by flux. We naturally think of spherical polar coordinates and we know that:

$$(\nabla^2 + k^2)e^{ikr} = 0$$

and we'd like to write the solutions in terms of j_l, n_l, and Y_{lm}.

$$e^{ikx} = \sum_{l=1}^{\infty} i^l(2l+1)j_l(kr)P_l(\cos\theta)$$

where $Y_{l0} = P_l(\cos\theta)$ with normalisation. This should be memorised if possible. Recalling that

$$j_l(\rho) \sim \frac{1}{r}\sin\Big(\rho - \frac{l\pi}{2}\Big) \qquad \& \qquad n_l(\rho) \sim (-1)\frac{1}{\rho}\cos\Big(\rho - \frac{l\pi}{2}\Big)$$

we have

$$e^{ikr} = \sum_{l=0}^{\infty} i^l(2l+1)P_l(\cos\theta)\frac{1}{2iR}\Big[\frac{e^{ikr} - il\pi/2}{r} - \frac{e^{-ikr} + il\pi/2}{r}\Big]$$

Where we have decomposed the free particle into both a incoming and outgoing wave. We latter learn that in elastic collsions (kinetic energy conserved) the scattering only depends on the outgoing section. Thus,

$$\psi(r) \sim \sum_{l=0}^{\infty} i^l(2l+1)P_l(\cos\theta)\Big(\frac{i}{2k}\Big)\frac{1}{\sqrt{V}}\Big[(-1)S_l(k)j_l^+(r) + j_l^-(r)\Big]$$

for large r and after the scattering occurs. We consider $S_l(k) = 1 + (S_l(k) - 1)$ where the latter corrsponds to the scattered particle.

$$\psi(r) = \frac{e^{ikr}}{\sqrt{V}} + \frac{1}{\sqrt{V}}\sum_{l=1}^{\infty}(2l+1)P_l(\cos\theta)\Big(\frac{S_l(k)-1}{2ik}\Big)\frac{e^{ikr}}{r}$$

we need to remember that we're doing all this in the COM frame. We call the latter function's coefficient $f_l(k)$ and,

$$\psi(r) = \frac{e^{ikr}}{\sqrt{V}} + f(\theta)\frac{e^{ikr}}{r}\frac{1}{\sqrt{V}}$$

by defining

$$f(\theta) = \sum_{l=0}^{\infty}(2l+1)P_l(\cos\theta)f_l(k)$$

Let's see probability flux conservation.

$$0 = \int J^i_{\text{net}}\, dA_i = \int (J_{\text{out}} + J_{\text{in}})^i\, dA_i =$$

provides us with the fact that $|S_l(k)| = 1$. The phase change of the scattering matrix element is given as

$$S_l(k) = e^{2i\delta_l}$$

where δ_l is called the scattering phase shift. Finding the flux,

$$j = \frac{\hbar}{2im}\Big(\psi^*\nabla\psi - \psi\nabla\psi^*\Big) = \frac{1}{V}\frac{\hbar k}{m} + \frac{1}{V}\frac{\hbar k}{m}\frac{|f(\theta)|^2}{r^2}\hat{r} + \dots$$

where m is the reduced mass. Using our defintion of the differential cross section, the numerator is:

$$r^2\, d\Omega\, \frac{1}{V}\frac{\hbar k}{m}\frac{|f(\theta)^2|}{r^2}$$

and dividing by flux $\hbar k/mV$, and we find

$$\frac{d\sigma}{d\Omega} = |f(\theta)|^2$$

Let's substitute $f(\theta)$!

$$\begin{aligned}
f(\theta) &= \sum_{l=0}^{\infty}(2l+1)f_l(k)P_l(\cos\theta) \\
&= \sum_{l=0}^{\infty}(2l+1)\Big(\frac{S_l(k)-1}{2ik}\Big)P_l(\cos\theta) \\
&= \sum_{l=0}^{\infty}(2l+1)e^{i\delta_l}\Big(\frac{\sin\delta_l}{k}\Big)P_l(\cos\theta)
\end{aligned}$$

for elastic collisions, and

$$\begin{aligned}\sigma_{\text{tot}} =& \int \frac{d\sigma}{d\Omega}\, d\Omega = 2\pi \int d(\cos\theta)\, |f(\theta)|^2 \\ =& 2\pi \int d(\cos\theta) \sum_{l,l'} (2l+1)(2l'+1) e^{i\delta_l - i\delta_{l'}} \Big(\frac{\sin\delta_l}{k}\Big)\Big(\frac{\sin\delta_{l'}}{k}\Big) P_l(\cos\theta) P_{l'}(\cos\theta) \\ =& \frac{4\pi}{k^2} \sum_l (2l+1)\sin^2\delta_l(k) = 4\pi \sum_l (2l+1)|f_l(k)|^2\end{aligned}$$

Corollary. Let's compute the imaginary part of $f(\theta = 0)$. We have:

$$\operatorname{Im} f(0) = \sum_l (2l+1)\frac{\sin^2\delta_l}{k^2} = \frac{k}{4\pi}\sigma_{\text{tot}}$$

This is called the optical theorem and is true in general.

Lecture 26 (December 8th)

Recall. Last class, we saw how in elastic colisions, where

$$S_l = e^{2i\delta_l}$$

But in inelastic collisions,

$$S_l = \eta_l e^{2i\delta_l}$$

where $0 \leq \eta_l \leq 1$.

Example. (Solid sphere) Consider a spherical potential well described by a pair $(-V_0, a)$. Recall that we could describe the free particle far from the potential with a combination of (j_l, η_l) which for large r became $(\sin r/r, \cos r/r)$. As we have finished the whole calculation for an arbitrary case, we only need to compute $S_l(k) = e^{2i\delta_l}$. In other words,

$$\delta_l = \delta_l(V_0, a)$$

We can do this through solving the Schrodinger equation for the cases $r \leq a$ and $r \geq a$, and impose that the solution and it's derivative is continuous at a. As the system is spherically symmetric, the solution is constructed of R_l and Y_{lm}, but we expect $E \geq 0$.

$$-\frac{\hbar^2}{2m_r}\Big(\frac{d^2}{dr^2} + \frac{2}{r}\frac{d}{dr} - \frac{l(l+1)}{r^2}\Big)R_l(r) + V(r)R_l(r) = ER_l(r)$$

Let $\kappa = 2m(E+V_0)/\hbar^2$. Solving,

$$\begin{cases} r \leq a & C_1 j_l(\kappa r) + C_2 n_l(\kappa r) \\ r \geq a & C_3 j_l(\kappa r) + C_4 n_l(\kappa r)\end{cases}$$

$C_2 \to 0$ due to divergence at zero, and continuity conditions give us:

$$\kappa \frac{j'(\kappa a)}{j_l(\kappa a)} = \kappa \frac{j_l'(\kappa a) + \dfrac{C}{B}\eta_l'(\kappa a)}{j_l(\kappa a) + \dfrac{C}{B}\eta_l(\kappa a)}$$

Notice all the potential information is in κ, and this fixes C/B.

$$\psi(r \geq a) = B\Big(j_l(kr) + \frac{C}{B}\eta_l(kr)\Big)$$

taking $r \to \infty$,

$$\psi(r \geq a) \sim B\Big(\frac{\sin(kr - l\pi/2)}{kr} - \frac{C}{B}\frac{\cos(kr - l\pi/2)}{kr}\Big)$$

and

$$\frac{2}{kr}\frac{1}{C/B - i}\Big[e^{-ikr - il\pi/2}\Big(\frac{-C/B + i}{C/B - i}\Big) - e^{-ikr + il\pi/2}\Big]$$

Telling us that

$$S_l(k) = e^{2i\delta_l} = \Big(\frac{-C/B + i}{C/B - i}\Big)$$

Solving this with respect to δ_l,

$$\tan\delta_l(k) = \frac{kj_l'(ka)j_l(\kappa a) - \kappa j_l(ka)j_l'(\kappa a)}{kn_l'(ka)j_l(\kappa a) - \kappa n_l(ka)j_l'(\kappa a)}$$

at $ka \ll 1$, we can series expand the above, and knowing that $ka \sim l$, we only need small orders of l. The central point is that when the denominator is zero, $\delta_l(k)$ becomes $(n\pi + \pi/2)$, and σ_l becoes maximum. We call this resonant scattering. We can approximate the above as

$$\tan\delta_l(k) = \gamma\frac{(ka)^{2l+1}}{E - E_k}$$

near resonance, and we can obtain $\sin^2\delta_l(k)$ which is linear to the cross section.

$$\sigma_l = \frac{4\pi}{k^2}(2l+1)\frac{(\Gamma/2)^2}{(E - E_k)^2 + (\Gamma/2)^2}$$

This is called the Breit-Wigner formula.

Example. (Direct solution of $l = 0$ or s wave) Again,

$$R_0(r \leq a) = C' j_0 = C'\frac{\sin\kappa r}{\kappa r}$$

and

$$R_0(r \geq a) = C_1 j_0 + C_2\eta_0 = C_1\frac{\sin kr}{kr} + C_2\Big(\frac{-\cos kr}{kr}\Big) = \frac{C''\sin(kr + \delta_0)}{kr}$$

This ultimately gives us

$$\Big(\frac{\sin(kr - l\pi/2 + \delta_l)}{kr}\Big)$$

for the radial equation and for $l = 0$, we find $\delta_0 = \delta_l(k)$.

$$\frac{1}{R_0}\frac{dR_0}{dr}\Big|_{R=a} \qquad \& \qquad \tan^{-1}\Big(\frac{k}{\kappa}\tan\kappa\Big) = ka$$

we see that for $V_0 > 0$, $\delta_0 > 0$, which makes sense as the differential cross section increases.

Theorem. (Born approximation) The Born approximation applies to:

(i) High energy

(ii) Weak potential

The idea is to express $d\sigma$, the transition rate into $d\Omega$ divided by incident flux p_i/Vm_r, and use, for the transition rate, the Fermi Golden rule. Recall,

$$\Gamma_{i\to f} = \frac{2\pi}{\hbar}\sum_f \delta(E_f - E_i)|\langle\phi_f|\lambda H^{(1)}|\phi_i\rangle|^2$$

We can also express the final state sum as as a phase integral,

$$\sum_f \to \int \frac{V\,d^3p}{(2\pi\hbar)^3}$$

where initial and final states are just free plane waves:

$$\frac{e^{ikr}}{\sqrt{V}}$$

notice how the the iner product in this case becomes a Fourier transform,

$$\int d^3r\, e^{-ik_f r}V(r)e^{ik_i r} = \tilde{V}(k_i - k_f)$$

we will ultimately find

$$\frac{d\sigma}{d\Omega} \propto |\tilde{V}(k_i - k_f)|^2$$